

5. Longworth traps are used to capture small mammals alive. They are baited with oats and supplied with a source of water. After being examined the mammals are released unharmed. Longworth traps must be checked at least once a day. **Image 5.1** shows a field vole and **Image 5.2** shows a Longworth trap.

Image 5.1



Field vole

Image 5.2



Longworth trap

Describe how you would estimate the population size of the field vole (*Microtus agrestis*) in a field of grass using the capture-recapture technique and Longworth traps. You have 10 Longworth traps set at the same coordinates in the field for the whole time of the investigation and a pair of small scissors to clip 1 cm² of hair from the backs of the voles.

Population size can be estimated using the following equation:

$$\text{Population size} = \frac{\text{number in 1}^{\text{st}} \text{ sample} \times \text{number in 2}^{\text{nd}} \text{ sample}}{\text{number in 2}^{\text{nd}} \text{ sample previously marked}}$$

(NB quadrats are not required in this investigation.)

[6 QER]

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